Lucas Swierad

401-536-0120 | lswierad@umich.edu | www.linkedin.com/in/lswierad00 |

Education

University of Michigan Ann Arbor, MI
Bachelor of Science in Computer Science & Bachelor of Science in Robotics *Aug. 2023 – May 2027*

Experience

Undergraduate Research Assistant Oct. 2026 – Present
UMich Robotic Exploration Lab *Ann Arbor, MI*

- Collaborated in a 4-person team to scope, specify, and integrate Chip-Scale Atomic Clock (CSAC) timing interfaces and acoustic hardware for One-Way Travel Time AUV navigation
- Evaluated low-frequency (500 Hz − 2 kHz) acoustic transducers and Popoto modems to support microsecond-accurate underwater range estimation
- Configured embedded microcomputers to acquire real-time acoustic payload data and support onboard signal processing algorithms for GPS-denied localization

Human-Robot Systems Teaching Assistant Jan. 2026 – Present
University of Michigan College of Engineering *Ann Arbor, MI*

- Led weekly laboratory and discussion sections for 60+ students, covering human-robot interaction, interface design, teleoperation, automation, situation awareness, and ethical robotics design
- Mentored student teams through the engineering design process, including stakeholder interviews, requirements development, design ideation, usability studies, risk assessment, and iterative prototyping

Site Reliability Engineering Intern June 2026 – Aug. 2026
Barracuda Networks *Ann Arbor, MI*

- Migrated 14 active CI/CD pipelines from Jenkins to GitHub Actions, enabling retirement of the organization's legacy Jenkins infrastructure
- Hardened Azure deployment workflows by using Terraform to provision managed identities and Azure Key Vault access for runtime secret retrieval, eliminating persistent credentials from CI/CD configuration

Projects

Self-Balancing Omnidirectional Ball-Bot | *Onshape, Python, PID Control, 3D Printing, Kinematics* Sep. 2026

- Designed and built a three-omni-wheel ballbot, including the mechanical assembly, 3D-printed components and kinematic mapping from body motion to individual wheel commands.
- Implemented a 100 Hz multithreaded control pipeline between a Raspberry Pi and RP2040, coordinating sensor acquisition, state estimation and motor actuation over LCM.
- Implemented and experimentally evaluated feedforward, PID and combined feedforward-plus-PID wheel-speed control using encoder feedback.

Inverse Kinematics Solver & RIX Service | *C++, Eigen, ROS-like Middleware (RIX)* Nov. 2025

- Implemented a C++ numerical Inverse Kinematics solver using Eigen to calculate geometric Jacobian pseudo-inverses and compute joint configurations via gradient descent
- Developed a high-performance C++ middleware service node (iksrv) to process asynchronous motion-planning requests and return joint configuration trajectories

Dynamic Scene Reconstruction in 4DGS | *PyTorch, 4DGS, GroundingDINO, SAM 2, NN* April 2026

- Extended the official 4D Gaussian Splatting implementation with a learned sigmoid-gated branch that separates dynamic and static deformation features
- Evaluated the method on D-NeRF and HyperNeRF sequences, measuring rendering quality with PSNR; improved PSNR by 0.25% relative to the baseline

Technical Skills

Programming Languages: Python, C++, C, Java, MATLAB, Julia, JavaScript, TypeScript, SQL
Frameworks: React, Node.js, Next.js, Flask, JUnit, WordPress, Cypress, Lint, FastAPI
Developer Tools: GIT/GitHub, Linux, Azure, AWS, Kubernetes, Docker, CI/CD (Jenkins + GitHub Actions), Postgres, MongoDB, Terraform, Jira, Ansible, VS Code
Systems & Middleware: ROS 2, LCM, Raspberry Pi, RP2040, Gazebo, TCP/IP Socket, Serialization
Relevant Coursework: ROS, Kinematics and Dynamics, Computer Vision, Deep Learning, Control Systems